

Integrated Rover Verification Plan

Create a verification plan that proves the rover meets its mission requirements without unnecessary testing.

Advanced**2 class periods****Skill Builder**

Engineering Context

Before final demonstration, the project manager needs a coordinated plan showing how each rover requirement will be inspected, analyzed, demonstrated, or tested.

What You Will Practice

- Map requirements to verification
- Sequence tests safely
- Define objective pass/fail criteria
- Plan data collection

What You Need

- Final requirement list
- Rover design information
- Verification matrix template
- Engineering notebook

Student Instructions

- 1 Number every mission requirement.
- 2 Choose inspection, analysis, demonstration, or test for each.
- 3 Write an objective pass/fail statement.
- 4 Identify equipment, setup, data, and responsible team role.
- 5 Arrange verification activities from low risk to high risk.
- 6 Identify requirements that can be verified in one combined test.
- 7 Create a retest rule for failed or inconclusive results.
- 8 Conduct a tabletop review with another team.

Engineering Requirements

- Every requirement has one or more verification methods
- Every method has a measurable pass/fail criterion
- Plan includes setup, equipment, and data to record
- Safety checks precede powered testing
- Retest criteria are defined

Constraints & Safety

- Do not schedule destructive tests before non-destructive evidence is collected
- Avoid using demonstration when quantitative data are required
- Identify assumptions and unavailable equipment

What You Submit

- Verification matrix
- Test sequence
- Data-recording plan
- Peer-review corrections
- Final readiness statement

Reflection Questions

- 1 Which requirement was most difficult to verify?
- 2 What tests could be combined without confusing results?
- 3 How will you handle an inconclusive result?
- 4 What requirement creates the greatest test risk?

Engineering Tip Test one requirement under one clearly controlled condition unless a combined test is deliberately planned.	Common Mistake “It worked” is not a pass/fail criterion. State the measured threshold and condition.
Stretch Goal Create a verification dashboard that automatically marks requirements complete.	Career Connection Verification and validation engineers plan evidence that proves aerospace systems meet requirements before delivery.